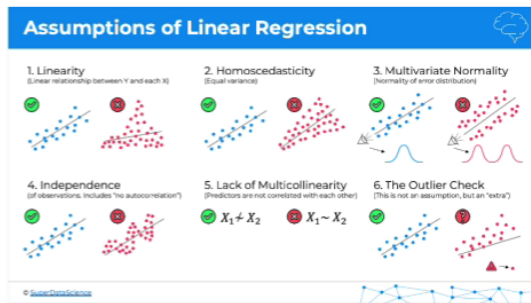


Notes on Linear Regression

Luis Mendez

I. ASSUMPTIONS OF LINEAR REGRESSION

- Linearity: The relationship between the independent and dependent variable is linear.
- Independence: The residuals (errors) are independent of each other.
- Homoscedasticity: The residuals have constant variance at every level of the independent variable.
- Normality: The residuals are normally distributed.
- Multivariate Normality: The independent variables are normally distributed (for multiple regression).
- No Multicollinearity: The independent variables are not highly correlated with each other.



II. BUILDING A MODEL

- Define the problem and collect data.
- Preprocess the data (handle missing values, encode categorical variables, etc.).
- Split the data into training and testing sets.
- Train the linear regression model on the training data.
- Evaluate the model using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), or R-squared on the testing data.
- Interpret the coefficients to understand the relationship between independent and dependent variables.